

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I, S. Susannal Devapriyam (192525398) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S. Susannal Devapriyam

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.
2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.
3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance

user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.
- 6.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

Question 1:

A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

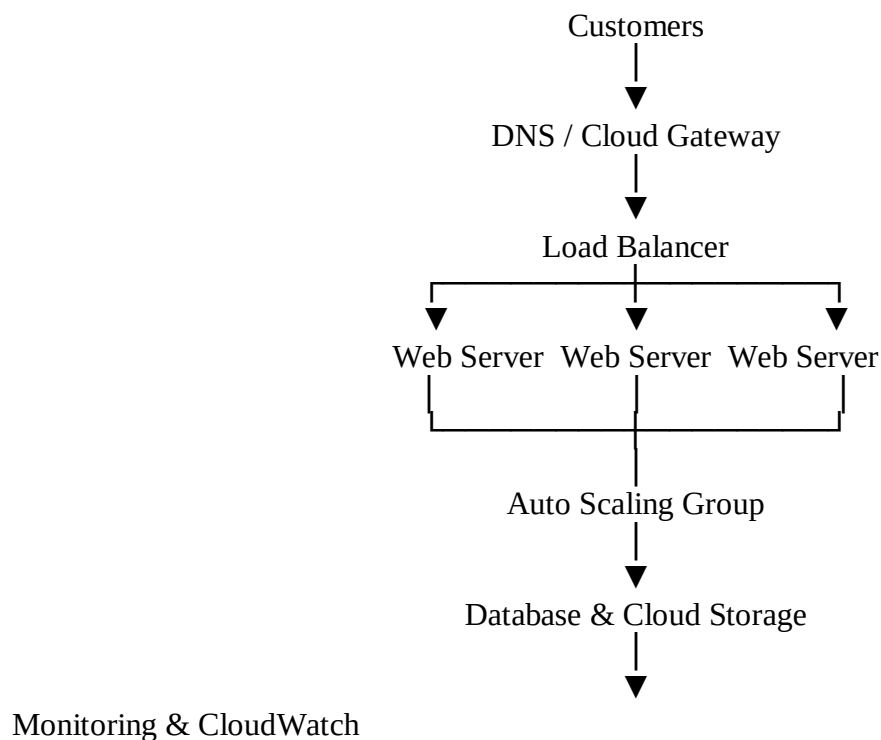
Answer:

Introduction

Many e-commerce companies experience sudden increases in website traffic during seasonal sales such as Black Friday, Diwali, Christmas, or Amazon Great Indian Festival. During these events, millions of customers visit the website at the same time to purchase products. If the system cannot handle this heavy traffic, the website may become slow, crash, or stop responding. This results in poor customer experience and financial loss.

Cloud computing provides services such as Auto Scaling, Load Balancing, and Monitoring to handle unpredictable traffic efficiently. These services ensure that the website remains available, performs well, and provides a smooth shopping experience even during peak demand.

Proposed Cloud Architecture



This architecture distributes user requests, automatically increases computing resources when demand grows, and continuously monitors system performance.

Role of Auto Scaling

Auto Scaling is a cloud service that automatically increases or decreases the number of servers according to user traffic.

During High Traffic

When thousands or millions of customers visit the website during a sale:

- New virtual servers are created automatically.
- The workload is shared among all servers.
- Website performance remains fast.

During Low Traffic

After the sale ends:

- Extra servers are removed automatically.
- Only the required servers continue running.
- The company pays only for the resources it uses.

Advantages of Auto Scaling

- Handles unexpected traffic spikes.
- Prevents server overload.
- Reduces manual effort.
- Improves application availability.
- Saves cloud costs.

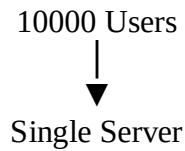
Role of Load Balancing

A Load Balancer distributes incoming customer requests equally among multiple servers.

Instead of sending every request to one server, it shares the workload.

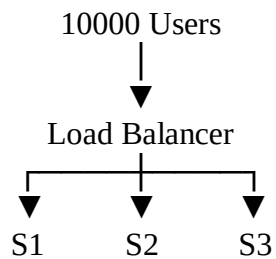
Example

Without Load Balancer



The server becomes overloaded and may crash.

With Load Balancer



Each server handles only a part of the total requests.

Benefits

- Better response time.
- Prevents server failure.
- High availability.
- Improved reliability.
- Efficient resource utilization.

How Auto Scaling and Load Balancing Work Together

Both technologies complement each other.

1. Customers send requests to the website.
2. The Load Balancer distributes requests across available servers.
3. Auto Scaling monitors traffic.
4. If traffic increases, new servers are automatically created.
5. If traffic decreases, unnecessary servers are removed.

This process happens automatically without human intervention.

Reducing Downtime

Downtime means the website becomes unavailable.

Cloud technologies reduce downtime by:

- Automatically replacing failed servers.
- Redirecting traffic to healthy servers.
- Creating additional servers during heavy demand.
- Preventing server overload.

As a result, customers can continue shopping without interruption.

Improving User Experience

A fast website increases customer satisfaction.

Using cloud technologies provides:

- Faster page loading.
- Smooth checkout process.
- Reduced waiting time.
- Minimal buffering for product images and videos.
- Continuous website availability.

Happy customers are more likely to complete their purchases.

Role of Monitoring Tools

Monitoring tools continuously observe cloud resources.

Popular monitoring tools include:

- AWS CloudWatch
- Azure Monitor
- Google Cloud Monitoring

These tools monitor:

- CPU usage
- Memory usage
- Network traffic
- Response time
- Server health
- Error rates

Benefits of Monitoring

- Detects problems early.
- Sends alerts to administrators.
- Helps Auto Scaling decide when to add or remove servers.
- Improves overall system performance.

Real-World Example

Consider **Amazon** during the **Great Indian Festival Sale**.

Millions of customers visit Amazon within a short time.

Amazon uses:

- Load Balancers to distribute customer requests.
- Auto Scaling to launch additional servers automatically.
- Monitoring tools to track server performance.
- Cloud storage to manage product information.
- CDN services to deliver images quickly.

Even when traffic increases dramatically, customers can continue shopping smoothly because cloud resources are automatically adjusted.

Advantages of This Cloud Solution

- Supports millions of users simultaneously.
- Prevents website crashes.
- Automatically handles traffic spikes.
- Reduces downtime.
- Improves customer satisfaction.
- Optimizes cloud costs.
- Provides high availability.
- Increases business revenue during seasonal sales.

Conclusion

A cloud-based architecture using Load Balancing, Auto Scaling, and Monitoring Tools is an ideal solution for companies facing unpredictable traffic spikes. Load balancing distributes customer requests efficiently, while auto scaling automatically adjusts resources according to demand. Monitoring tools continuously observe system health and trigger scaling whenever required. Real-world companies such as Amazon successfully use these technologies to ensure smooth shopping experiences during major sales events. This cloud architecture improves performance, reduces downtime, optimizes costs, and enhances customer satisfaction.

Question 2

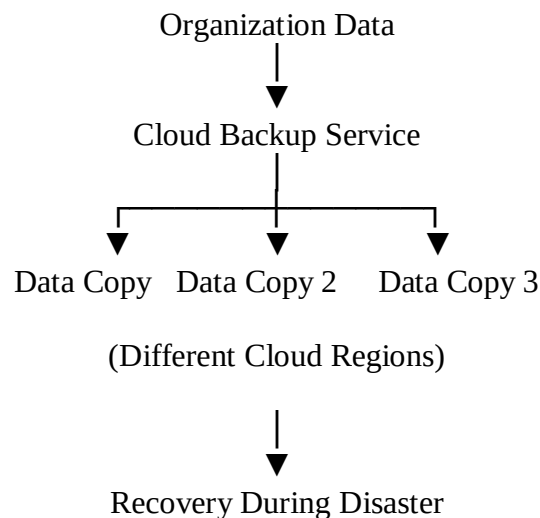
An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.

Answer:

Introduction

Every organization generates large amounts of important data, including customer information, financial records, employee details, and business documents. Losing this data due to hardware failure, cyberattacks, or accidental deletion can seriously affect business operations. Cloud storage provides a secure and reliable solution for storing backup data while ensuring that it remains available even during unexpected failures.

Cloud Backup Architecture



This architecture ensures that multiple copies of the same data are stored safely in different locations.

Durability and Data Redundancy

Durability

Durability means that data remains safe and available for a very long time. Cloud providers automatically create multiple copies of files so that data is not lost even if one storage device fails.

For example, if one hard disk becomes damaged, another copy of the data is still available.

Data Redundancy

Data redundancy means storing identical copies of data in different servers or cloud regions.

Benefits include:

- Prevents permanent data loss.
- Provides continuous access to information.
- Improves reliability during hardware failures.

Encryption and Access Control

Cloud providers protect backup data using strong security mechanisms.

Encryption

Encryption converts readable data into unreadable coded information.

Two types are used:

Data at Rest

- Protects files stored in cloud storage.

Data in Transit

- Protects data while it is transferred between users and the cloud using secure protocols like SSL/TLS.

Access Control

Access control ensures that only authorized users can access backup files.

Common methods include:

- Username and Password
- Multi-Factor Authentication (MFA)
- Role-Based Access Control (RBAC)

These methods reduce unauthorized access and improve security.

Cost Advantages of Cloud Storage

Compared to traditional storage systems, cloud storage offers many financial benefits.

- No need to purchase expensive storage hardware.
- No maintenance cost for physical servers.
- Pay only for the storage actually used.
- Easy to increase or decrease storage capacity.
- Reduced electricity and cooling costs.

This makes cloud storage more affordable for organizations.

Disaster Recovery Use Case

Consider a hospital that stores patient medical records.

If the hospital server is damaged due to a flood or fire:

- Backup copies stored in the cloud remain safe.
- Data is restored quickly.
- Doctors can continue accessing patient records without major interruption.

This ensures continuous healthcare services.

Advantages

- High durability.
- Data redundancy.
- Strong security through encryption.
- Lower storage cost.
- Easy disaster recovery.
- High availability.
- Automatic backup.

Conclusion

Cloud storage provides secure, reliable, and cost-effective backup solutions. Features such as durability, redundancy, encryption, and access control protect valuable business data. During disasters, cloud backups help organizations recover quickly and continue business operations with minimal downtime.

Question 3

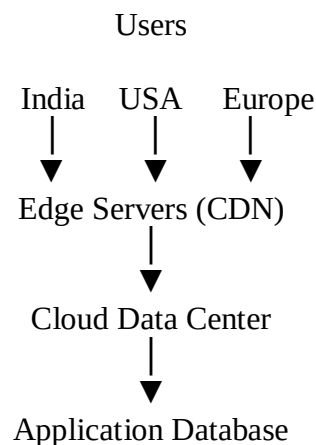
A mobile app company plans to deploy its application globally. Explain how Content Delivery Networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

Answer:

Introduction

A mobile application used worldwide must provide fast response times for users in different countries. If all users access a single server located in one region, they may experience delays and slow performance. Cloud providers solve this problem using Content Delivery Networks (CDNs) and edge servers.

Global Cloud Architecture



This architecture delivers content from the nearest server to each user.

Content Delivery Network (CDN)

A CDN is a network of servers located around the world.

Instead of sending content from one central server, the CDN delivers content from the nearest available server.

Benefits

- Faster loading time.
- Reduced network traffic.
- Lower latency.

- Better application performance.
- Reduced server workload.

Edge Servers

Edge servers are located close to users in different geographical regions.

When a user opens the application:

- The request is sent to the nearest edge server.
- Cached content is delivered immediately.
- Users experience faster response times.

Advantages

- Quick access to content.
- Less buffering.
- Better user experience.
- Reduced delays.

Global Availability

Cloud providers such as AWS, Microsoft Azure, and Google Cloud operate data centers in many countries.

They provide:

- Multiple cloud regions.
- Availability Zones.
- Automatic traffic routing.
- Disaster recovery.
- Global load balancing.

These services ensure that applications remain available even if one region experiences problems.

Real-World Example

Consider **Netflix**.

Millions of users watch movies from different countries.

Netflix uses:

- CDNs.
- Edge servers.
- Cloud infrastructure.

As a result:

- Videos load quickly.
- Buffering is reduced.
- High-quality streaming is maintained worldwide.

Similarly, online games such as PUBG and Fortnite use edge servers to reduce network delay and improve gameplay.

Advantages

- Low latency.
- High-speed content delivery.
- Better customer experience.
- Reduced bandwidth usage.
- High global availability.
- Improved application performance.

Conclusion

CDNs and edge servers help mobile applications provide fast and reliable services to users across the world. By using cloud providers with global infrastructure, organizations can deliver content efficiently while reducing latency and improving user satisfaction.

Question 4

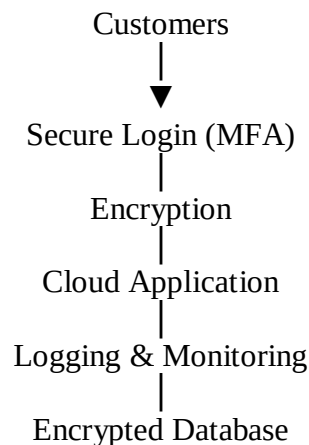
A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.

Answer:

Introduction

Financial institutions handle highly sensitive information such as customer bank accounts, credit card details, loan records, and transaction history. Protecting this information is extremely important because any security breach may lead to financial loss and legal consequences. Cloud providers offer various security services to help organizations meet compliance and regulatory requirements.

Secure Cloud Architecture



This architecture protects financial information from unauthorized access.

Regulatory Compliance and Auditing

Cloud providers support compliance by following international security standards such as:

- ISO 27001
- PCI DSS
- SOC 2
- GDPR (where applicable)

They also maintain audit logs that record every user activity.

Auditing helps organizations:

- Track system access.
- Detect suspicious activities.
- Meet government regulations.
- Generate compliance reports.

Role of Logging

Logging records important system activities.

Examples:

- User login time.
- Transaction history.
- File access.
- System changes.

Logs help identify unauthorized activities and support security investigations.

Role of Monitoring

Monitoring continuously checks:

- CPU usage.
- Network traffic.
- Login attempts.
- Application health.
- Security alerts.

Monitoring tools automatically notify administrators if suspicious behavior is detected.

Role of Encryption

Encryption protects financial information while storing and transmitting data.

Data at Rest

Protects stored customer information.

Data in Transit

Protects online banking transactions between customers and servers using SSL/TLS encryption.

Data Sovereignty

Some countries require customer data to remain inside national borders.

Cloud providers support this by allowing organizations to:

- Store data in specific cloud regions.
- Restrict cross-border data transfer.
- Select regional data centers.
- Apply local compliance policies.

This helps organizations satisfy government regulations.

Real-World Example

Consider an online banking application.

When a customer transfers money:

1. The user logs in using Multi-Factor Authentication.
2. The transaction is encrypted.
3. Every activity is recorded in audit logs.
4. Monitoring tools detect suspicious login attempts.
5. Customer data remains stored in approved national cloud regions.

This ensures secure financial transactions and compliance with banking regulations.

Advantages

- Strong security.
- Regulatory compliance.
- Secure financial transactions.
- Continuous monitoring.
- Better customer trust.
- Protection against cyber attacks.
- Easy auditing.

Conclusion

Cloud providers help financial institutions protect sensitive information through encryption, logging, monitoring, auditing, and regulatory compliance. By following security standards and implementing data sovereignty policies, organizations can safely use cloud computing while protecting customer financial information and maintaining trust.

Question 5

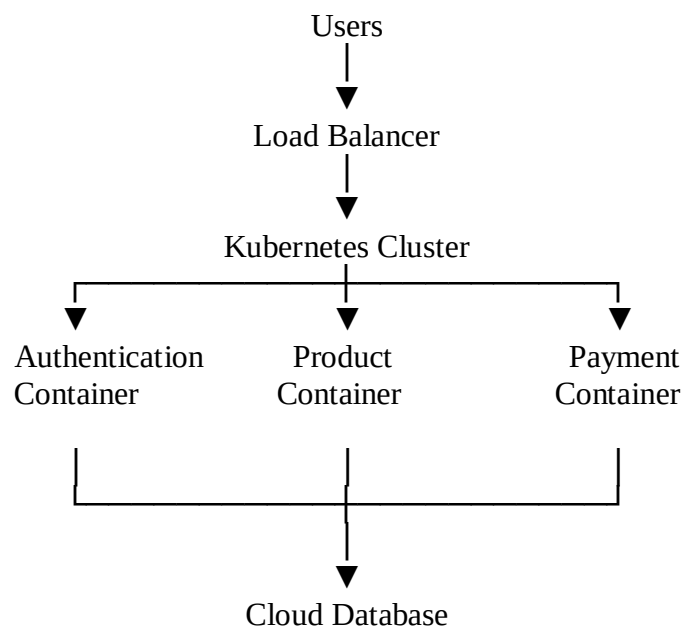
A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Answer:

Introduction

Modern software applications require fast deployment, easy maintenance, and the ability to scale according to user demand. Traditional Virtual Machines (VMs) consume more resources because each VM includes a complete operating system. Containers have become a popular alternative because they are lightweight, faster, and use fewer system resources. Container orchestration tools such as **Kubernetes** further simplify application deployment, management, and scaling across different cloud platforms.

Container-Based Architecture



In this architecture, each application service runs inside its own container and is managed by Kubernetes.

Containers vs Virtual Machines

Containers and Virtual Machines are both used to run applications, but they differ in performance and resource usage.

Containers	Virtual Machines (VMs)
Share the host operating system	Each VM has its own operating system
Lightweight	Heavyweight
Start within seconds	Take several minutes to start
Use less CPU, RAM, and storage	Require more CPU, RAM, and storage
Higher performance	Comparatively slower
Suitable for cloud-native applications	Suitable for running different operating systems

Performance

Containers start much faster because they do not need to boot an entire operating system. This improves application deployment speed and allows developers to release updates quickly.

Resource Usage

Since containers share the host operating system, they consume fewer resources compared to virtual machines. This enables multiple containers to run on the same server efficiently.

Role of Kubernetes in Deployment and Scaling

Managing hundreds of containers manually is difficult. Kubernetes automates this process.

Deployment

Kubernetes automatically deploys application containers across available servers.

Scaling

If user traffic increases:

- Kubernetes creates additional container instances automatically.

If traffic decreases:

- Extra containers are removed automatically.

Self-Healing

If a container crashes:

- Kubernetes automatically restarts or replaces the failed container.

Load Balancing

Kubernetes distributes incoming requests among multiple containers, ensuring smooth application performance.

Portability Across Different Cloud Environments

One major advantage of containers is **portability**.

A container created on one cloud platform can run on another cloud platform without modifying the application.

For example, the same container can run on:

- Amazon Web Services (AWS)
- Microsoft Azure
- Google Cloud Platform (GCP)
- Private Cloud
- Hybrid Cloud

This reduces vendor dependency and makes application migration easier.

Example of a Microservices-Based Application

Consider an **Online Food Delivery Application**.

Instead of building one large application, it is divided into smaller services.

Example:

- **User Service** – Manages user accounts.
- **Restaurant Service** – Displays restaurant information.
- **Order Service** – Processes food orders.
- **Payment Service** – Handles online payments.
- **Delivery Service** – Tracks delivery status.
- **Notification Service** – Sends SMS and email updates.

Each service runs in its own container.

If the **Payment Service** receives heavy traffic, Kubernetes increases only the number of payment containers without affecting the other services.

This improves performance and reduces resource usage.

Advantages of Containerization

- Faster application deployment.
- Better performance.
- Lower resource usage.
- Easy scalability.
- Automatic recovery using Kubernetes.
- High portability across cloud providers.
- Supports microservices architecture.
- Reduces infrastructure cost.

Real-World Example

A company like **Netflix** uses a microservices architecture where different services such as user authentication, video streaming, recommendations, and billing run independently. Containers package these services, while Kubernetes manages deployment, scaling, and recovery. During peak viewing hours, Kubernetes automatically creates additional streaming containers to handle increased demand, ensuring uninterrupted service.

Conclusion

Containerization provides a lightweight and efficient method for deploying modern applications. Compared to virtual machines, containers offer better performance, lower resource usage, and faster startup times. Kubernetes simplifies deployment by automating container management, scaling, load balancing, and recovery. The portability of containers allows applications to run across different cloud environments without modification. Combined with microservices architecture, containerization helps organizations build scalable, reliable, and cost-effective cloud applications.